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| **PI Name:** | Blakely, Eleanor A Ph.D. | | |
| **Project Title:** | Early Markers of Space Radiation Induced Cataractogenesis | | |
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| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Radiation health | | |
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| **PI Email:** | eablakely@lbl.gov | **Fax:** | FY 510-486-4475 |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 510-486-6595 |
| **Organization Name:** | Lawrence Berkeley National Laboratory | | |
| **PI Address 1:** | One Cyclotron Road | | |
| **PI Address 2:** | MS977 | | |
| **PI Web Page:** | | | |
| **City:** | Berkeley | **State:** | CA |
| **Zip Code:** | 94720 | **Congressional District:** | 13 |
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Cataract incidence in astronauts has been correlated with exposure to relatively low absorbed doses of radiation in space flight. The type of space radiation linked with low-dose cataractogenesis has not yet been identified. The goal of this proposal is to test the hypothesis that radiation-induced alterations in early molecular signaling in cultured human lens epithelial (HLE) cells can be used to identify early markers of relative risk of cataracts from key high-LET radiation components. We have shown that exposure to low fluences of high-LET radiation promotes severe lens cell dysfunction via events that include modulation of FGF-2, inhibition of cell cycle control via cyclin-dependent-kinase inhibitors (CDKIs) altered expression of cell receptor and adhesion molecules, and altered apoptotic/enucleation program machinery, in the HLE and human lens fiber (HLF) cells. These observations are consistent with a model in which ionizing radiation initiates premature differentiation. We propose three specific aims that will use a matrix of charged particles of high energy and atomic number to test important aspects of our hypothesis, the link to cataractogenesis, and that will also allow the development of biological countermeasures. Quantitative measurements of the inappropriate

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| **Task Description:** | commencement of fiber cell differentiation will be made as a function of particle fluence, atomic number and energy, and correlated to the kinetics of early signaling expression in HLE cells in vitro, and in the nascent radiation-induced cataract in the rat in vivo. The human lens is one of the most radiosensitive organs of the body, and is one of the tissues upon which shielding requirements for space flight have been based to avoid cataracts, a late-appearing radiation effect. Recent evidence however indicates that exposure to relatively low doses of space radiation is associated with an increased incidence and earlier appearance of human cataracts than was anticipated. We developed an in vitro model of differentiating human lens epithelial (HLE) cells, and have been studying their response to different types of ionizing radiations prevalent in space travel under conditions that mimicked the normal growth environment in the tissue, but were more amenable to investigation than in vivo models. The objective of our project is to provide evidence showing that particle radiation induces alterations in gene expression profiles when compared to the normal lens differentiation process resulting in premature differentiation subsequently leading to cataract formation. Historically, radiation-induced cataract has been anatomically characterized as posterior sub-capsular (PSC) due to the location of its appearance within the lens after radiation exposure. The mechanisms underlying radiation-induced cataract are thought to involve direct DNA damage to the epithelial cells, which is later expressed as an opacification when the epithelial cells have differentiated into fiber cells and migrated to the PSC location. However lens fiber cells at the juncture of the posterior capsule and suture are terminally differentiated in the region where fibers cells are devoid of nuclear organelles including the cell nucleus. The radiation-induced cataract is a progressive lesion that continues to evolve during the life of the organism, and the molecular basis for the origin and development of radiation-induced cataract are largely unknown. The continuous lifetime migration of germinative epithelial cells down the anterior surface of the lens of the eye, and the progressive transition into primary and secondary fiber cells involves a coordinated symphony of changes. The results from our work will provide information on the important roles of several radiation-responsive signal transduction pathways in lens differentiation and of pre-cataractous changes. Understanding of the molecular basis for radiation-induced cataract pathology will provide important tools to develop biological approaches for countermeasures. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Our cell culture model is an exciting tool that demonstrates human lens epithelial cell growth on an extracellular matrix derived from bovine corneal endothelial cells, and also exhibits aspects of fiber cell differentiation with the expression of several morphological and protein markers. The results of our work will be used to pursue countermeasures that could reduce the risk of cataract. These studies may also contribute to estimations of risk assessment for radiation-induced cataracts among astronauts exposed to particles during space travel, especially during missions into deep space. We may also contribute to the development of countermeasures for radiation-induced cataracts from other occupational and clinical radiation exposures. The biological mechanisms underlying normal lens fiber cell differentiation may also be elucidated. The exact factor(s) responsible for the continued replication of lens epithelial cells in embryos and adults remains to be determined. These studies therefore should shed new light on the process of human lens cell differentiation, that has been difficult to explore with animal models in vivo or with human lens cultures in vitro that are either immortalized and/or undergo only limited differentiation. |
| **Task Progress:** | To understand the molecular basis of radiation-induced cataract, it is important to identify those genes with altered expression or function due to radiation exposure. Regulation of cell differentiation is dependent on the interplay among integrins, extracellular matrix ligands and cytoskeletal elements. We have used microarray analyses to investigate changes in gene expression in cultured human lens cells exposed to single acute doses of Xrays, 55 MeV/amu protons, or accelerated iron ions of either 1 GeV/amu or 600 MeV/amu. A cultured human lens cell model was used for these studies. Lens cells were grown on matrix-coated plastic tissue culture dishes and irradiated with 55 MeV/amu proton beams at Berkeley Lab, or 1 GeV/amu iron ions at the NASA Space Radiation Lab at Brookhaven National Lab. Xray control studies were completed at each institution. Single doses of either 0.5, 2, or 4 Gy were given. RNA and protein were harvested at different times from 2 hr to 12 hr after radiation exposure. For gene array analysis, total RNA from an unirradiated control and an irradiated sample were reverse transcribed, PCR amplified and processed per directions provided by the manufacturer (Superarray Biosciences, Corp, MD). The biotin-labeled PCR amplicons were hybridized to functional extracellular matrix (ECM) gene profiles, and applied to ECL film for fluorescent signal detection. Background noise signals were subtracted from the signal intensities for each gene spot and normalized to the positive housekeeping controls in each filter. Replicate experiments were completed. The expression of specific genes was quantitatively evaluated using real time RT-PCR. In a separate experiment, Sprague Dawley rats were either sham irradiated or exposed to 10 or 100cGy of 600MeV/u Iron ions or 1 GeV/u Titanium ions. Ophthalmological examinations are conducted using a digital video slit lamp image capture system. Using the ECM functional gene array approach, we have identified changes in the expression of a number of genes in unirradiated controls as cells differentiate from lens epithelial cells into fiber cells. Shortly after a single 2 Gy proton dose, a predominant number of proton-responsive genes were down-regulated in the differentiating fiber cells while others were upregulated in epithelial cells. Dose-dependent changes were also observed as evidenced by the fact that genes upregulated after a 2 Gy exposure were different from those after 4 Gy. Comparisons of ECM gene expression profiles after protons and iron ions show that radiation quality plays a role in the regulation of cell adhesion molecules in lens epithelial cells. Quantitative analysis of integrin alpha5 and connexin 50 confirmed that upregulation of these ECM genes is temporally regulated and dependent on radiation quality. As normal lens cells differentiate from epithelial cells into fiber cells, the regulation of expression of specific ECM genes are important in modeling the architecture of fiber cell alignment. We have demonstrated in our studies that exposure to ionizing radiation from proton or iron ion beams triggers the expression of a select profile of ECM genes in cultured human lens epithelial cells which may contribute to premature and aberrant differentiation. Western analyses are in progress to probe for translational changes in the expression of the candidate gene products. The data presented here provide functional evidence for radiation-induced premature defective differentiation supporting the previously published morphological evidence.<br>Sprague Dawley rats were exposed to an acute low dose of 600 MeV/amu Iron ions or 1 GeV/amu Titanium ions in April 2004 at the NSRLin BNL. Manual ophthalmologic examinations were performed by a veterinary ophthalmologist using a manual slit lamp instrument on fully-dilated animals. In addition, video recordings of cataract progression were obtained using the system developed by John Clark (University of Washington). All animals exposed to 100cGy of Iron or Titanium beams had scorable cataracts at 1 year after irradiation. Observable changes included stellate spoke-like lens fiber extensions. At necropsy, 14 months after exposure, lenses were extracted from the globe and ex vivo photographs of each lens were taken. Lenses were fixed, embedded and maintained at -80C until processing. We plan to section these samples to probe for the expression of specific molecular markers during the next funding period.<br><br>In order to strengthen the statistical power of our study, another cohort of animals were exposed to 10 cGy or 100 cGy of |

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| | 600 MeV/u Iron ions in the Spring of ’05. The in-life portion of this study is still in progress. Due to beam availability, we are planning to expose animals to Titanium ions in the Fall of ’05. |
| **Bibliography Type:** | Description: (Last Updated: 02/12/2026) |
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| **Presentation** | Blakely, E.A.; Chang, P.Y.; Bjornstad, K.A. and Rosen, C.J. "Differentatial Gene Expression induced in Human Lens Cells by Space Radiations" 16th Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, NY<br>May-2005 |
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